

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Townsend's Solitaire"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

```
[1] "divergent transitions: 0"
```

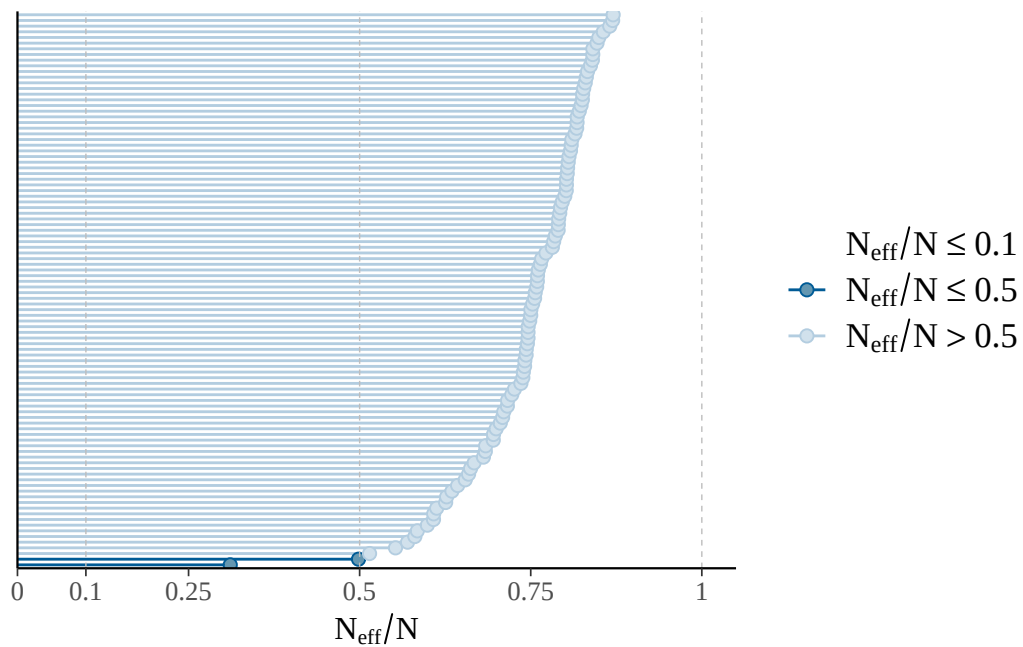
Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of `n_eff` to `N` (number of samples), the better.

Warning: The ESS has been capped to avoid unstable estimates.



```
[1] "effective sample size fine"
```

Warning: The ESS has been capped to avoid unstable estimates.

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5514	1.0664	1.2053	1.3669	1.4889	3.6021

[1] "n_eff ratio tail ESS:"

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4983	0.7135	0.7604	0.7514	0.8083	0.8704

[1] "effective sample size in bulk and tail fine"

Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.





Post-warmup iteration

Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9990	0.9995	0.9998	0.9999	1.0002	1.0024

```
[1] "All R-hat values < 1.01"
```

Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_bio1_3yrs"          "b_occ_sum_annual_crops_3yrs"
```

```
[1] "b_colo_bio1"              "b_colo_pr_spring"
```

```
[3] "b_colo_pr_winter"         "b_colo_Ibio7E2"
```

```
[5] "b_colo_sum_annual_crops"
```

```
[1] "b_ex_Ibio1E2"             "b_ex_Ibio2E2"          "b_ex_Ibio3E2"
```

```
[4] "b_ex_Ipr_springE2"        "b_ex_Ipr_winterE2"     "b_ex_sum_annual_crops"
```

